Data Analysis and Visualization Using R

Agrinformatics

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MS in GPB

Expertise:

- Plant Abiotic Stress
- Molecular Breeding
- Plant Bioinformatics
- R Programming

Code of Conduct

Theory – Basic Stats

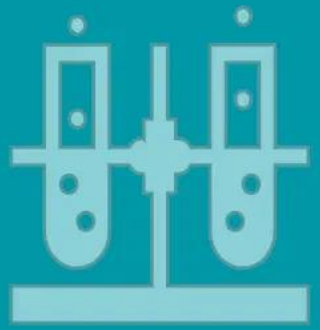
Descriptive vs Inferential Statistics

Descriptive and inferential statistics are two main branches of statistics that are used to analyze and interpret data.

DESCRIPTIVE

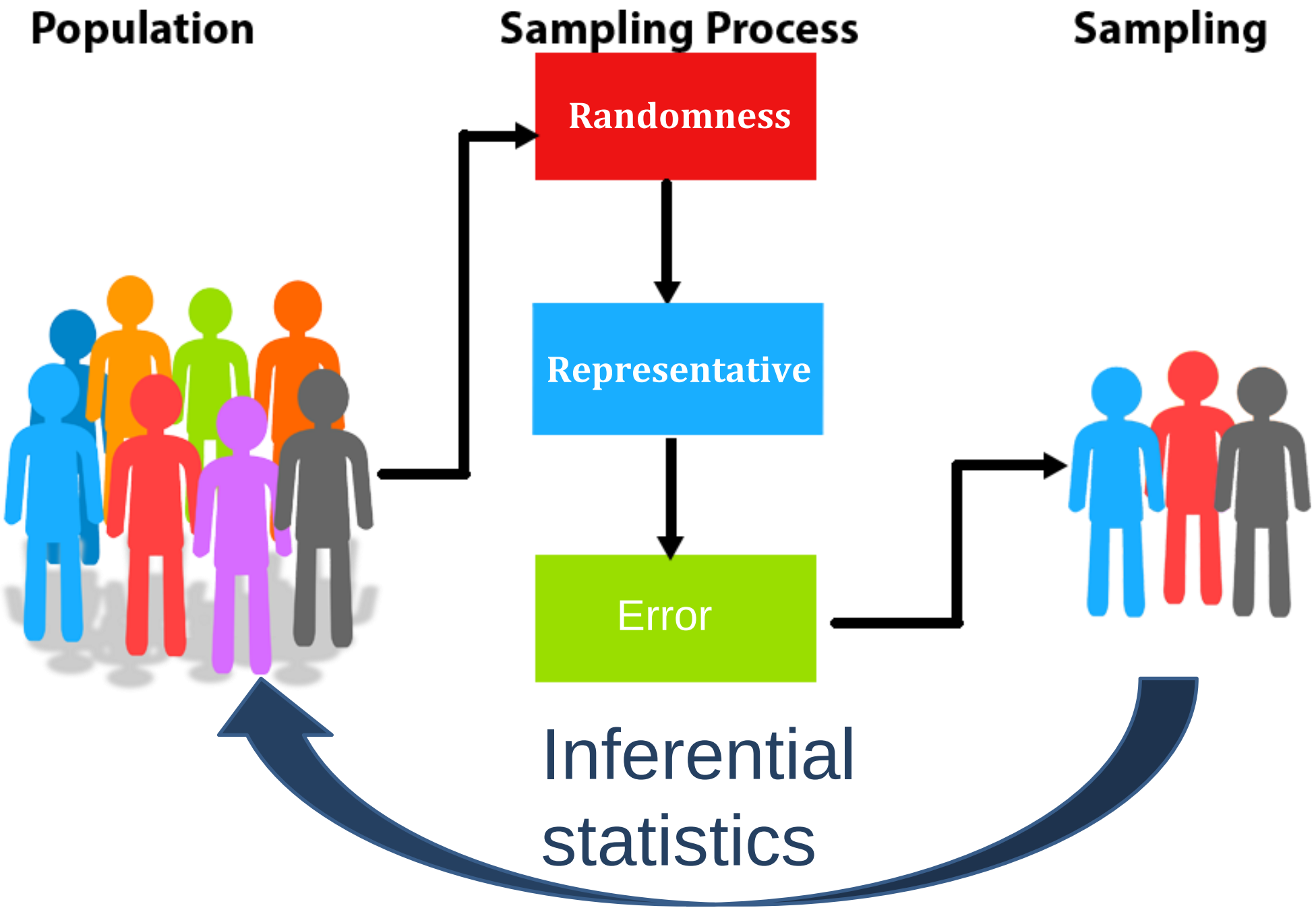
Descriptive statistics is a branch of statistics used to summarize and describe the characteristics of a dataset. Descriptive statistics involves calculating summary measures, such as the mean, median, mode, range, standard deviation, variance.

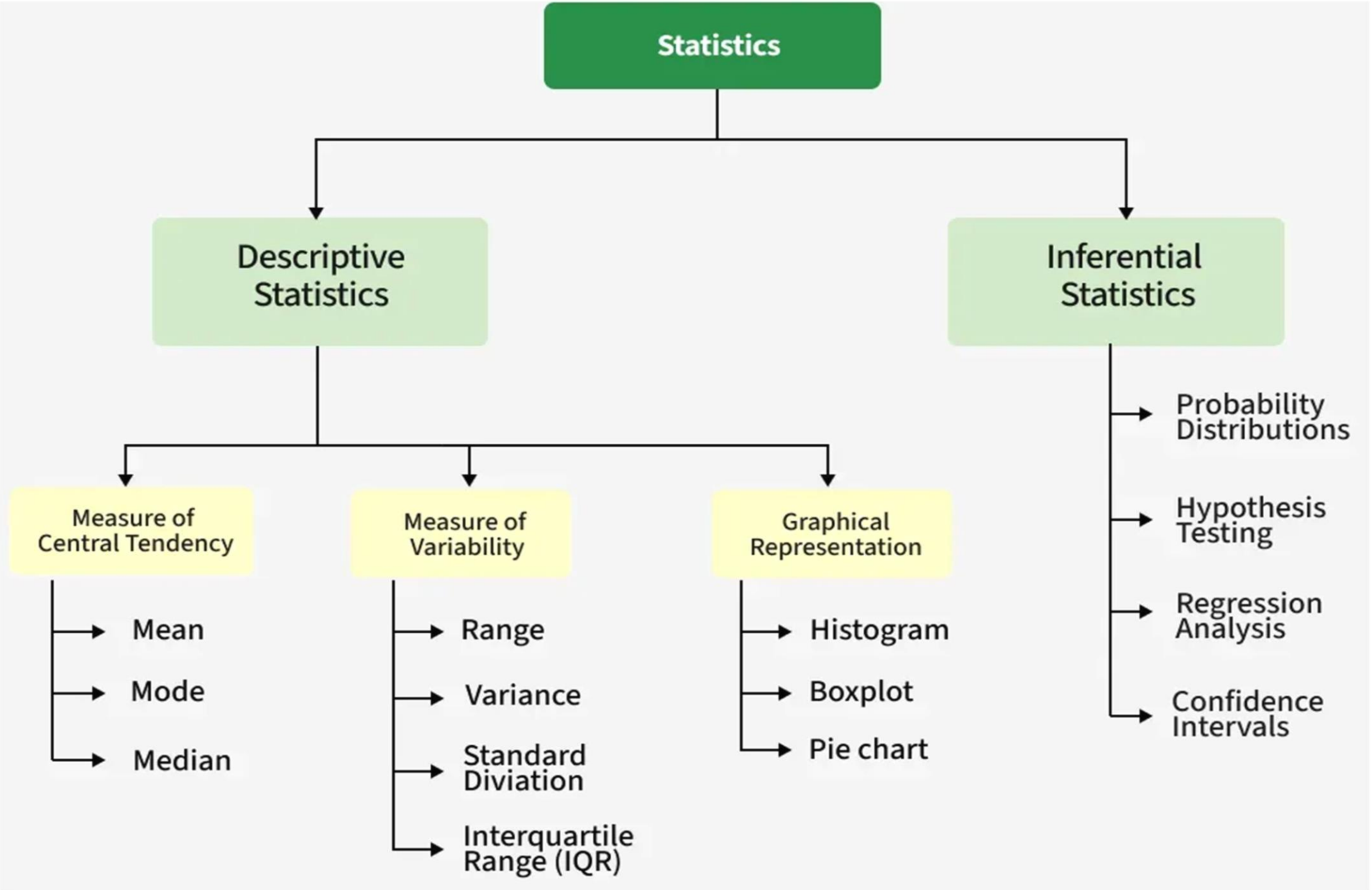
VS



INFERENTIAL

Inferential statistics is a branch of statistics used to make inferences or predictions about a population based on a sample of data. Inferential statistics involves using statistical tests, such as hypothesis tests and regression analysis.


Population
Sampling Process
Sampling
Randomness
Representative
Error
Inferential statistics


Statistics
Descriptive Statistics
Inferential Statistics
Measure of Central Tendency
Mean
Mode
Median
Measure of Variability
Range
Variance
Standard Diviation
Interquartile Range (IQR)
Graphical Representation
Histogram
Boxplot
Pie chart
Probability Distributions
Hypothesis Testing
Regression Analysis
Confidence Intervals

1 mean

Arithmetic average

1.5, 1.8, 2.0, 1.6, 1.7, 1.9, 2.1, 1.4, 1.8, 2.2

$$\frac{1.5+1.8....+2.2}{10} = 1.8 \text{ kg}$$

2 median

Middle value of ordered data

2, 8, 4, 30, 5, 10, 7, 3, 6, 5

Ord: 2, 3, 4, 5, **5**, **6**, 7, 8, 10, 30

(5 + 6) / 2 = **5.5 days**

3 mode

Most frequent value

Cough Fever Cough Discharge Fever Cough Cough Fever Fever Cough Cough Discharge Cough Cough Lameness Fever Fever Fever Cough Discharge

Cough = 9

Fever = 7

Discharge = 3

Lameness = 1

4 range

Difference between the highest and lowest values.

30, 32, 35, 37, 40, 42, 43, 45, 48, 49, 50, 58

Range = Max − Min = 58 − 30 = **28**

Standard deviation is a measure of how spread out the values in a dataset are from the mean.

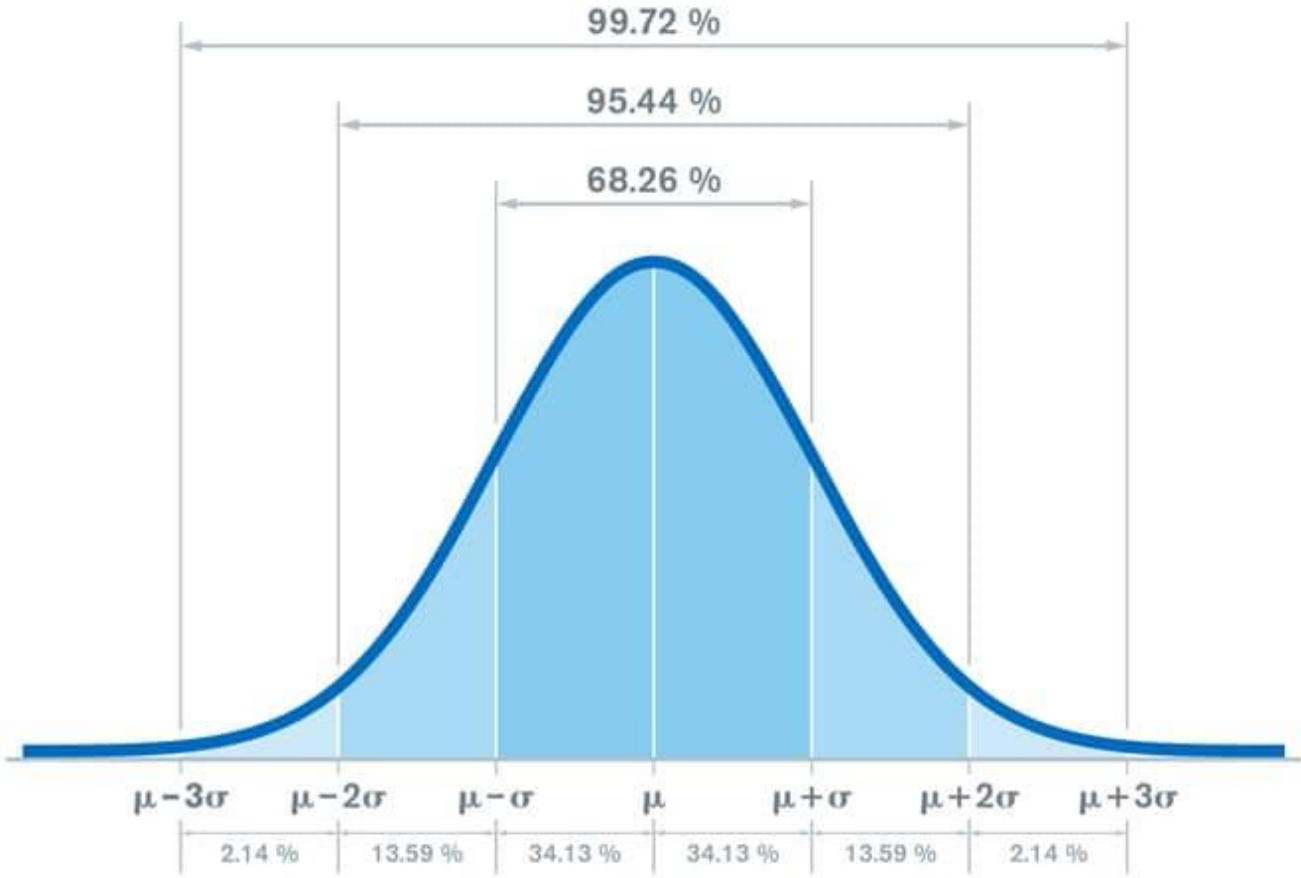
Population	Sample
$\sigma = \sqrt{\frac{\Sigma(x_i-\mu)^2}{n}}$	$S = \sqrt{\frac{\Sigma(x_i-\bar{x})^2}{n-1}}$
μ - Population Average xi - Individual Population Value n - Total Number of Population	x - Sample Average x_i - Individual Population Value n - Total Number of Sample

x	$\bar{x}$	$x - \bar{x}$	$(x-\bar{x})^2$
1	2.83	1 – 2.83 = (-1.83)	(-1.83)^2 = 3.35
2	2.83	2 – 2.83 = (-0.83)	(-0.83)^2 = 0.69
2	2.83	2 – 2.83 = (-0.83)	(-0.83)^2 = 0.69
3	2.83	3 – 2.83 = (0.17)	(0.17)^2 = 0.03
4	2.83	4 – 2.83 = (1.17)	(1.17)^2 = 1.37
5	2.83	5 – 2.83 = (2.17)	(2.17)^2 = 4.71

3.35 + 0.69 + 0.69 + 0.03 + 1.37 + 4.71 = 10.84

$$s^2 = \frac{10.84}{6-1} = 2.17$$

Mean±SD= 10.84±2.17



Quartile Deviation

6, 20, 4, 5, 3, 6, 7, 8, 4 Total Observations (n) = **9** **Given Data (Ordered):** 3, 4, 4, 5, 6, 6, 7, 8, 20

Range= 20−3 = 17 days **SD= 5.2 day**

Step 1:

Q1 position =1(n+1)/4
= 2.5th
Q3 position = 3(n+1)/4
= 7.5th

Step 2:

Q1 = 4 + 0.5 × (4 - 4)
= 4.0
Q3 = 7 + 0.5 × (8 - 7) =
7.5

Step 3:

QD = (Q3 - Q1) / 2
= (7.5 - 4.0) / 2
= 1.75 days

Thumb Rule

Quartile Deviation	Interpretation
< 1.0	Very low spread (high uniformity)
1.0 – 2.0	**Moderate spread** (moderate variability)
> 2.0	High spread (more variability)

Coefficient of Variation (CV)

$$\frac{\text{Standard Deviation (s)}}{\text{Sample Mean } (\bar{x})}$$

2. Assess Experimental Precision

Analyte: Glucose	Analyte: Glucose
Method: Hexokinase	Method: Glucose oxidase
Standard deviation = 4.8	Standard deviation = 4.0
Mean = 120	Mean = 100

Which method is more precise?

Calculate the CV to see:

$$CV\ (\%) = \frac{\text{Standard Deviation}}{\text{Mean}} \times 100$$

$$\frac{4.8\ (SD)}{120\ (Mean)} \times 100 = 0.04\%\ (CV) \qquad \frac{4.0\ (SD)}{100\ (Mean)} \times 100 = 0.04\%\ (CV)$$

1. Comparing Datasets with Different Units

Group	Glucose (mg/dL)	Glucose (mmol/L)	**mean**	**SD**	**CV%**
Sample 1	85	4.7	**90 mg/Dl**	**4.08 mg/Dl**	**4.53mg/Dl**
Sample 2	90	5.0			
Sample 3	95	5.3	**5 mmol/L**	**2.45 mmol/L**	**4.9 mmol/L**

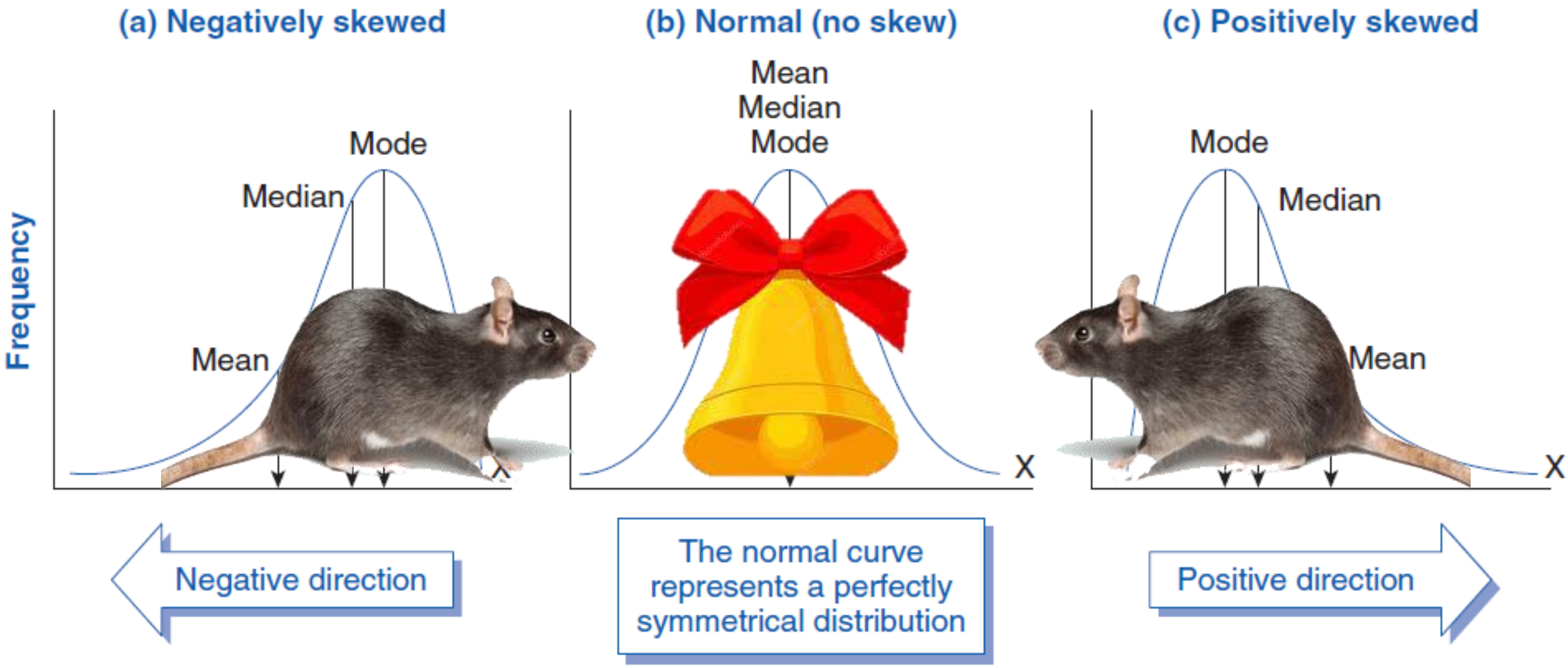
3. Stability Analysis in Crop Breeding

Crop Variety	Mean Yield (kg/plot)	CV (%)	Interpretation
Variety A	5.0	30%	High yield, but very **variable**
Variety B	3.5	10%	Moderate yield, but **very stable**

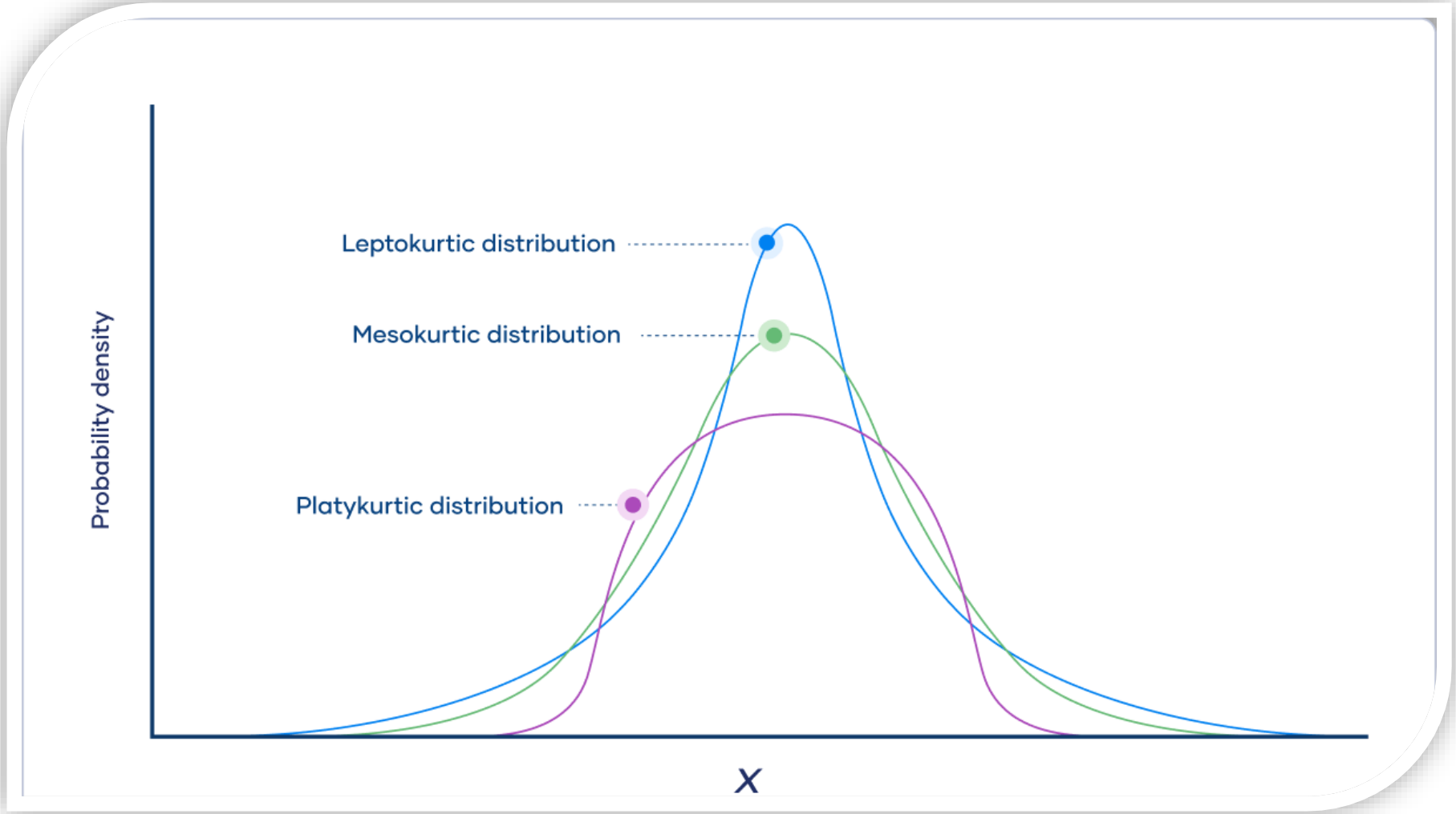
4. Data set with different mean

Metric	Dataset A (1,2,3)	Dataset B (100,101,103)
Mean	2	101.33
SD	1	1.53
CV (%)	50%	1.51%

Skewness is a measure of the asymmetry of a distribution.



Kurtosis is a measure of the tailedness of a **distribution**



Correlation is the relationship showing how two variables change together.

Pearson correlation (r)

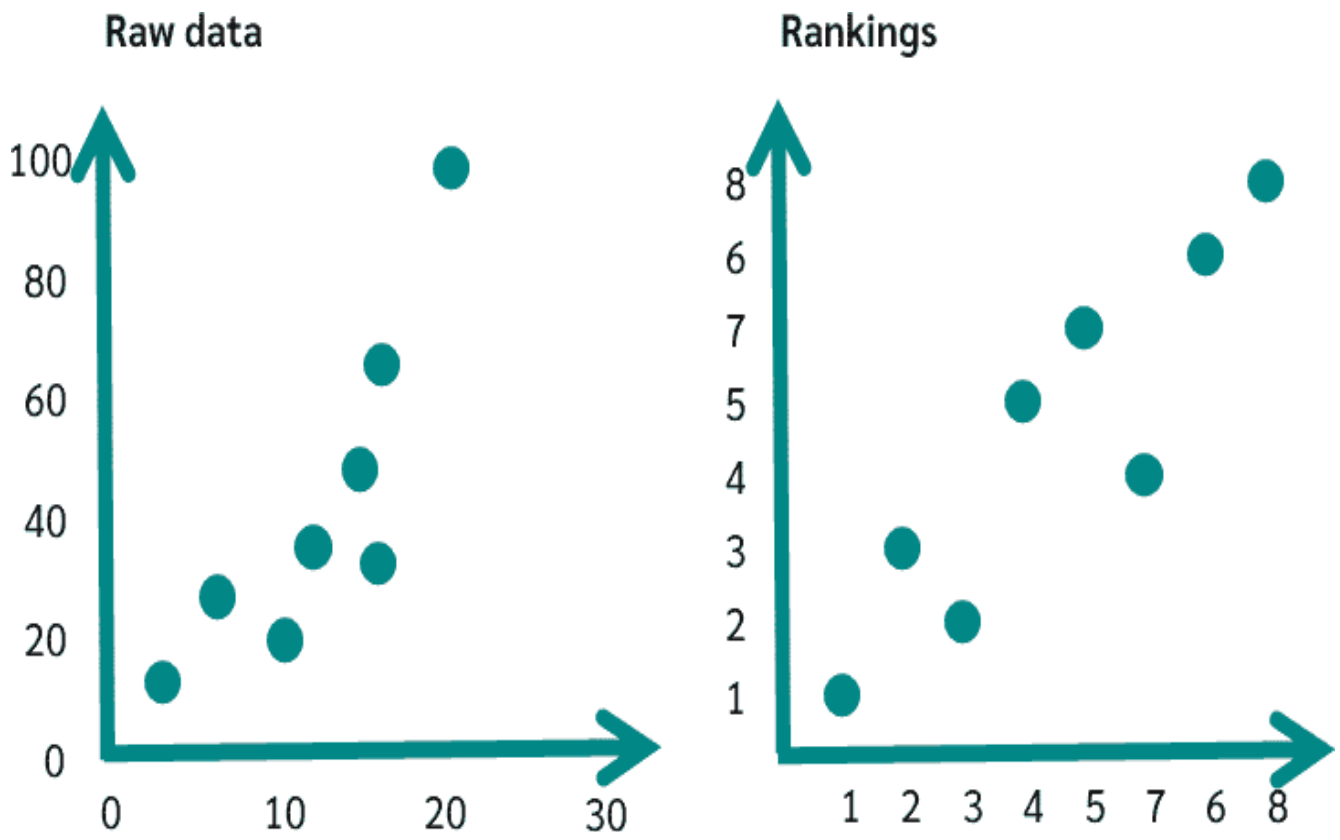
a measure of the strength of a linear association between two variables

Assumption

- Two metric variables
- Normally distributed

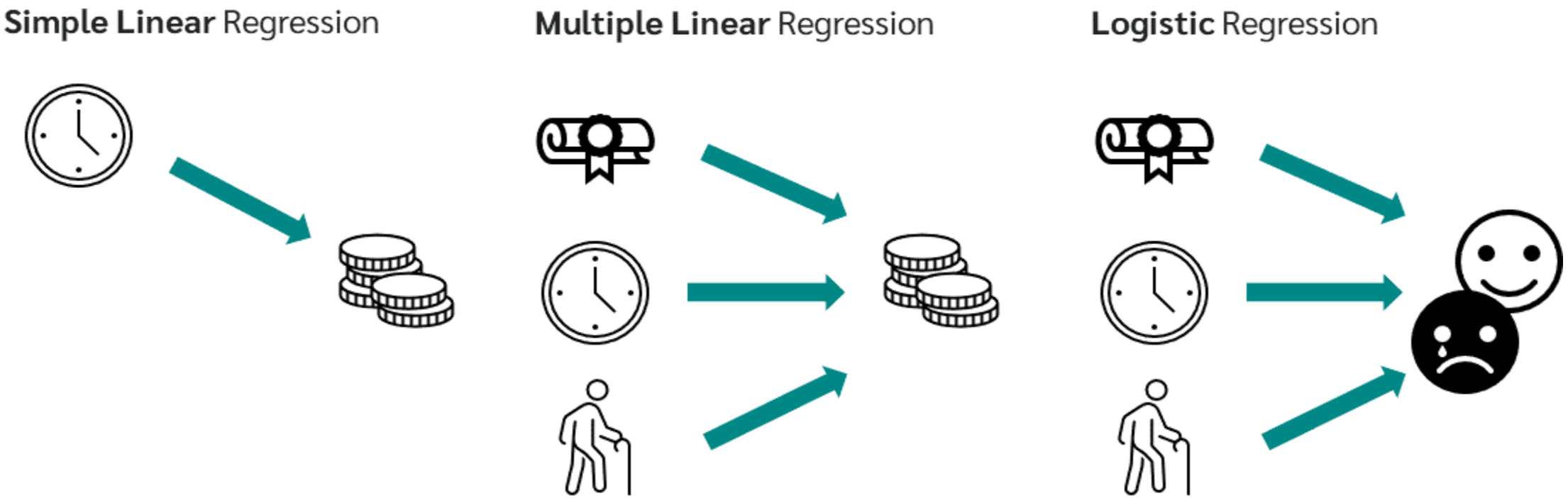
r value	Interpretation
r = 1	Perfect positive linear correlation
0.8 ≤ **r** < 1	Strong positive linear correlation
0.4 ≤ **r** < 0.8	Moderate positive linear correlation
0 < **r** < 0.4	Weak positive linear correlation
r = 0	No correlation
-0.4 ≤ **r** < 0	Weak negative linear correlation
0.8 ≤ **r** < -0.4	Moderate negative linear correlation
-1 < **r** < 0.8	Strong negative linear correlation
r = -1	Perfect negative linear correlation

Spearman Correlation (r_s)

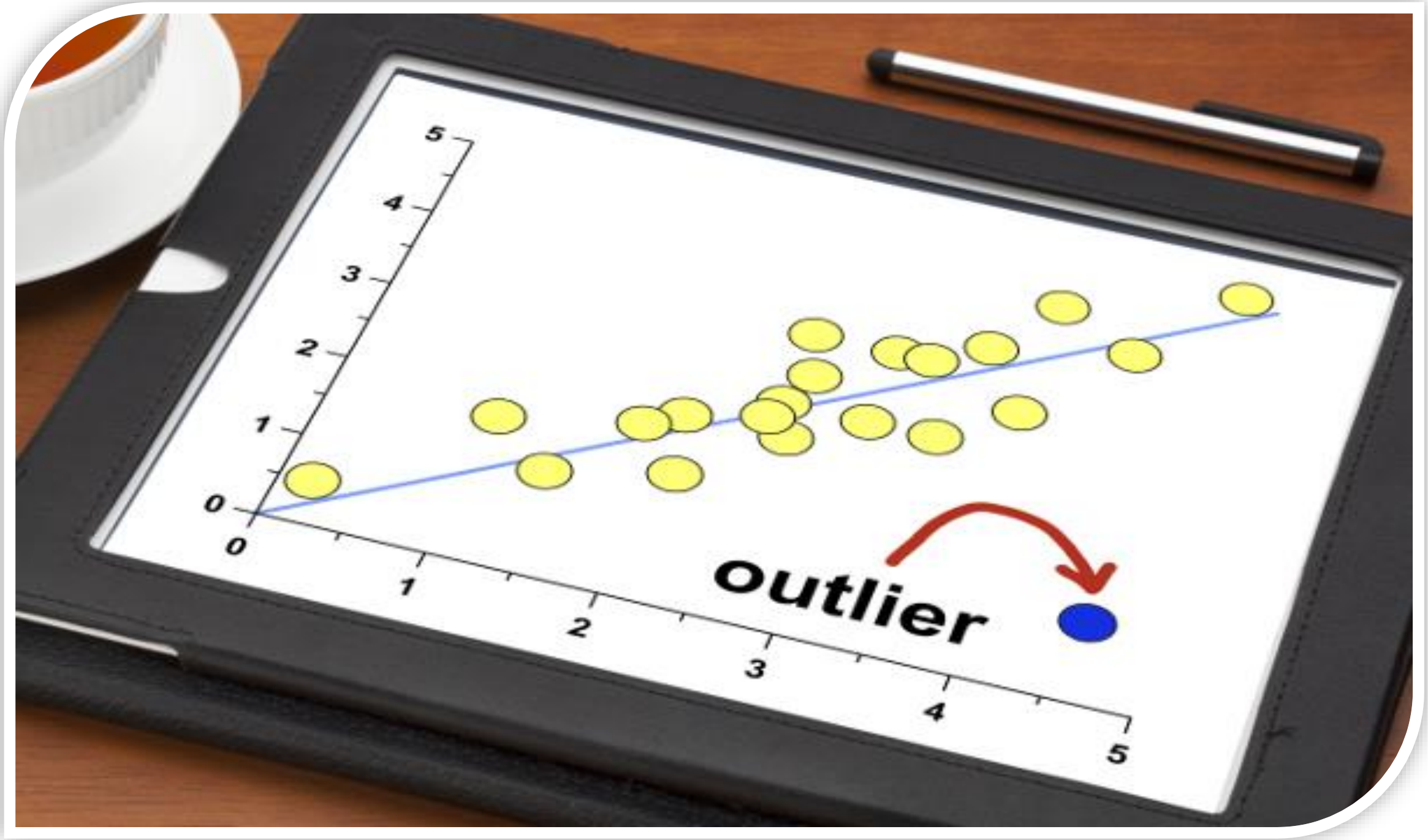


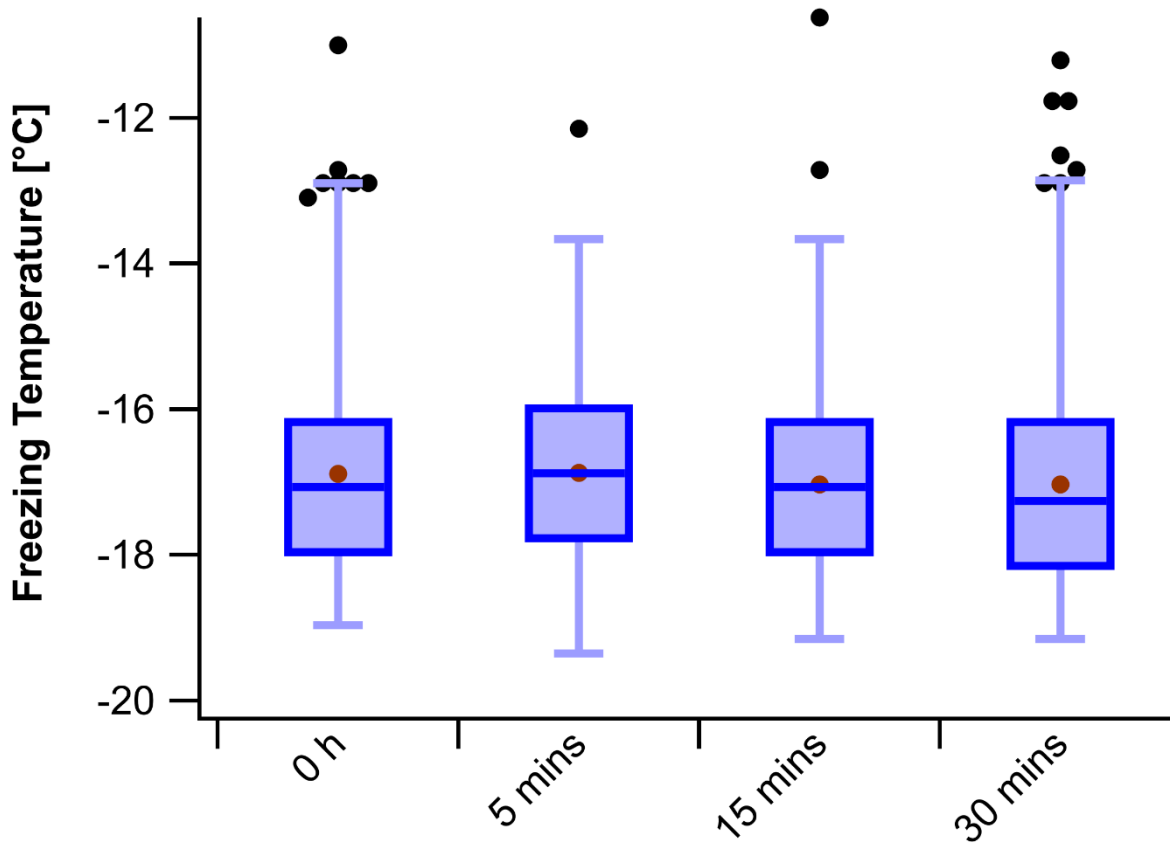
Regression analysis

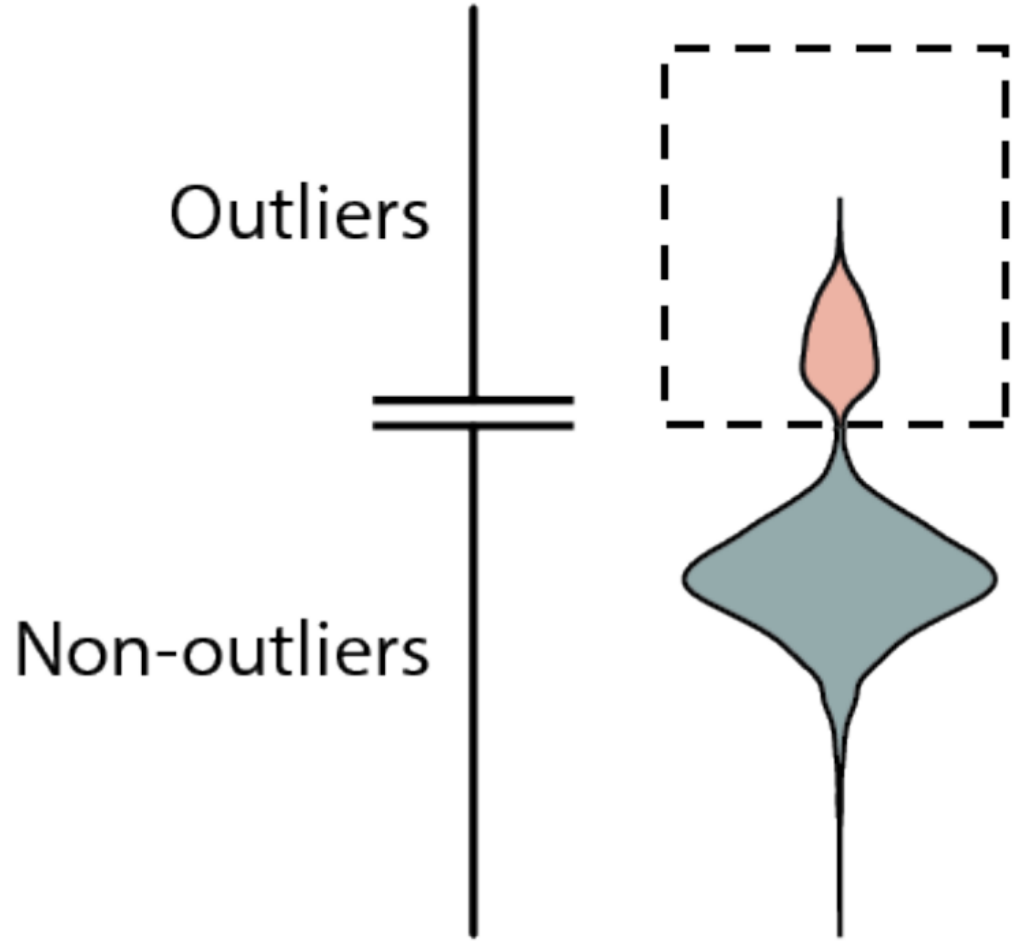
Relationships between a dependent variable and one or more independent variables



Regression Type	No. of Independent Variables	Dependent Variable Scale	Independent Variable Scale
Simple Linear Regression	One	Metric	Metric, Ordinal, Nominal
Multiple Linear Regression	Multiple	Metric	Metric, Ordinal, Nominal
Logistic Regression	One or Multiple	Ordinal, Nominal	Metric, Ordinal, Nominal

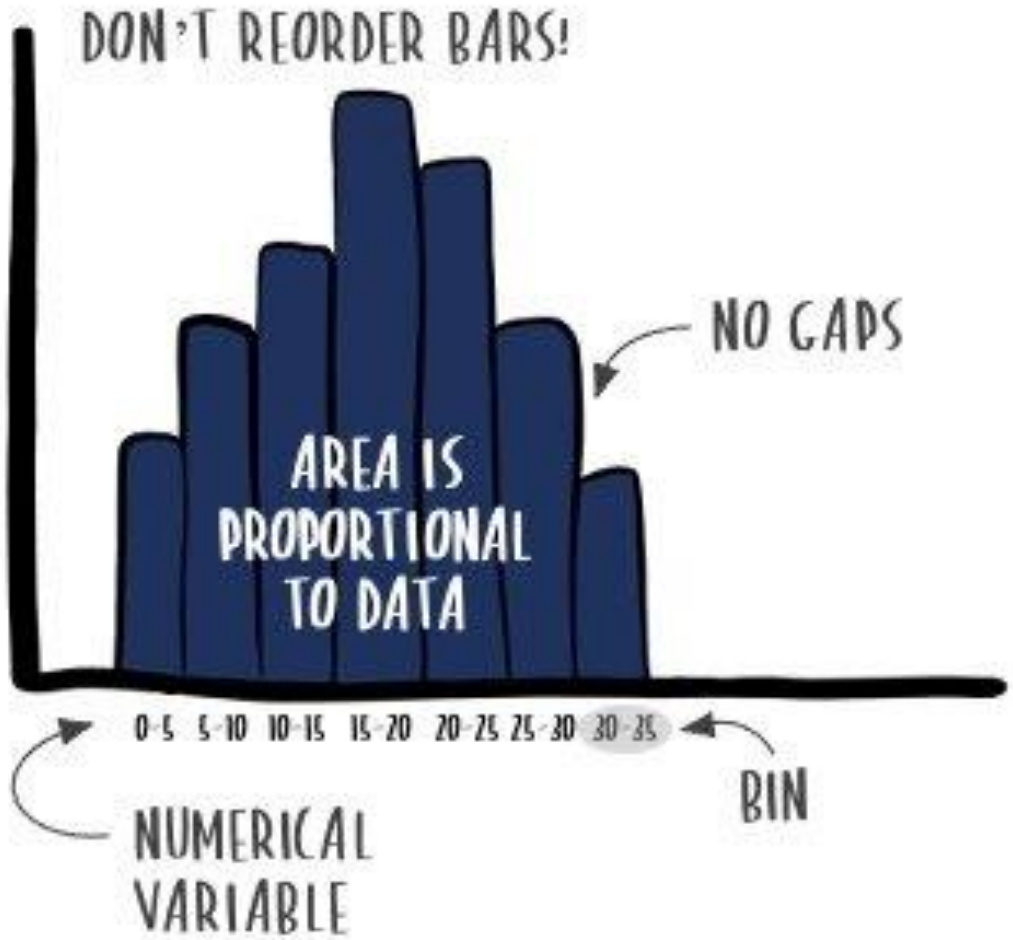

5
4
3
2
1
0
0
1
2
3
4
5
Outlier


Freezing Temperature [°C]
-12
-14
-16
-18
-20
0 h
5 mins
15 mins
30 mins

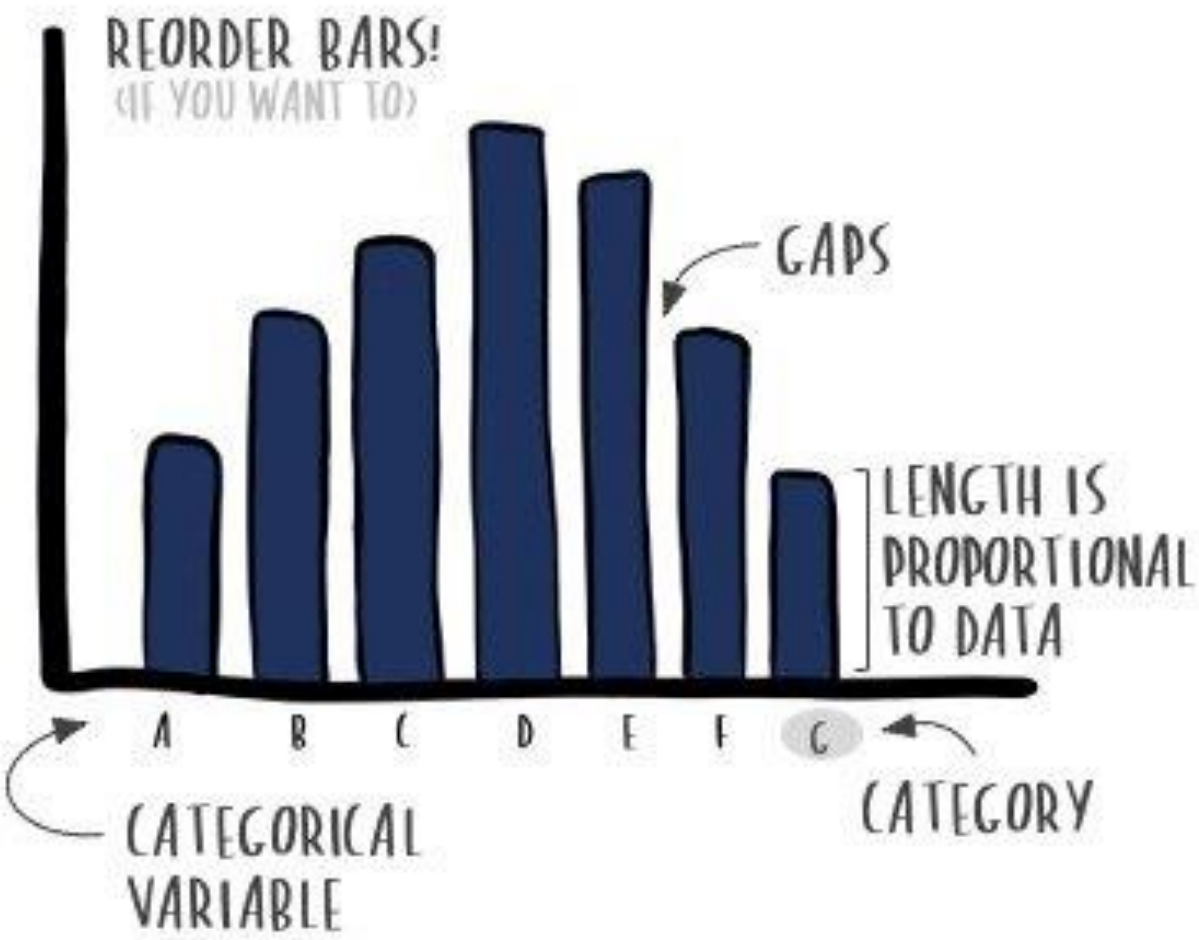

Outliers
Non-outliers

Don't let variability, outliers & skewness hide: ban bar graphs

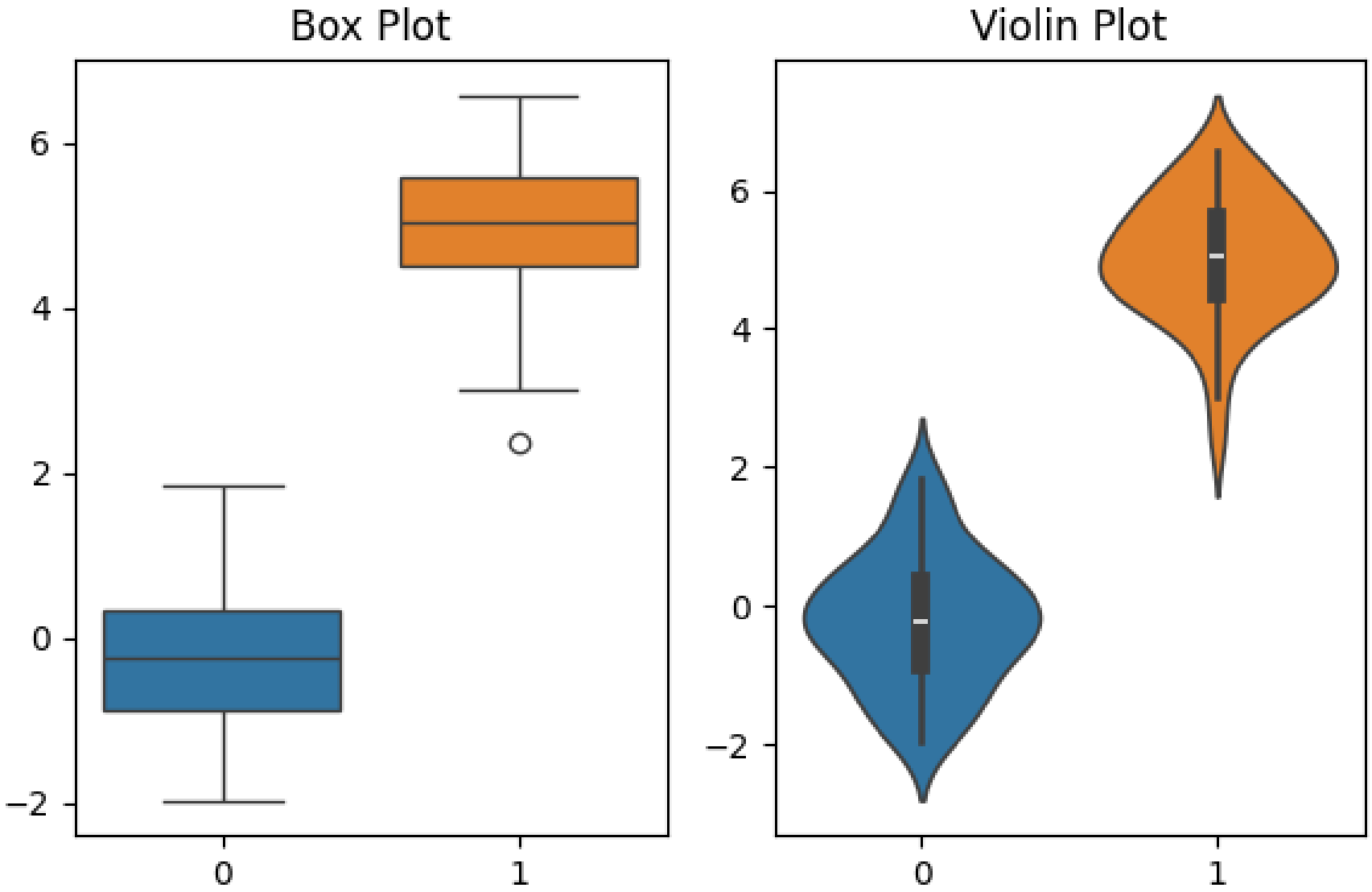
This is a **histogram**...

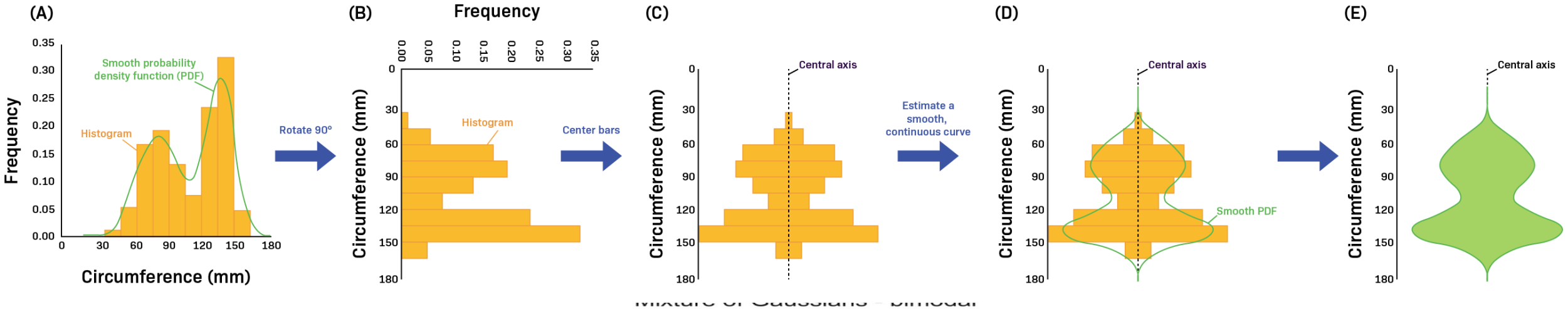


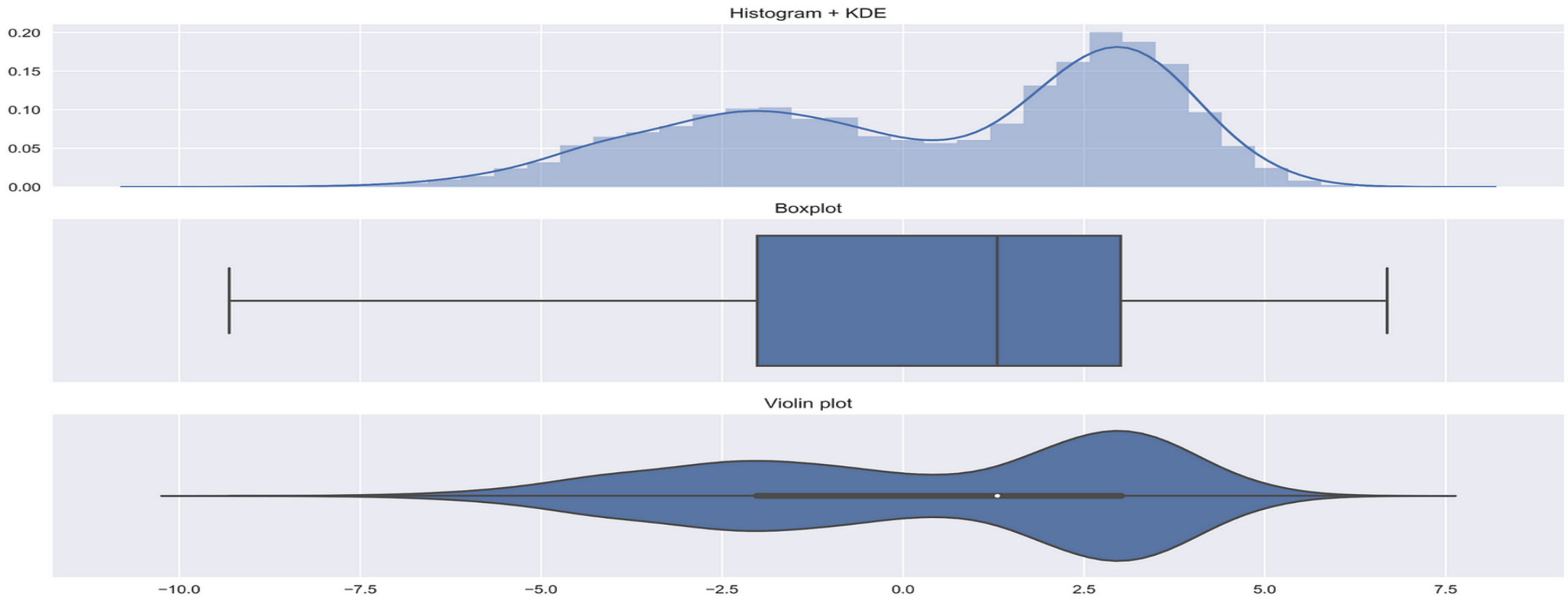
This is a **bar chart**...

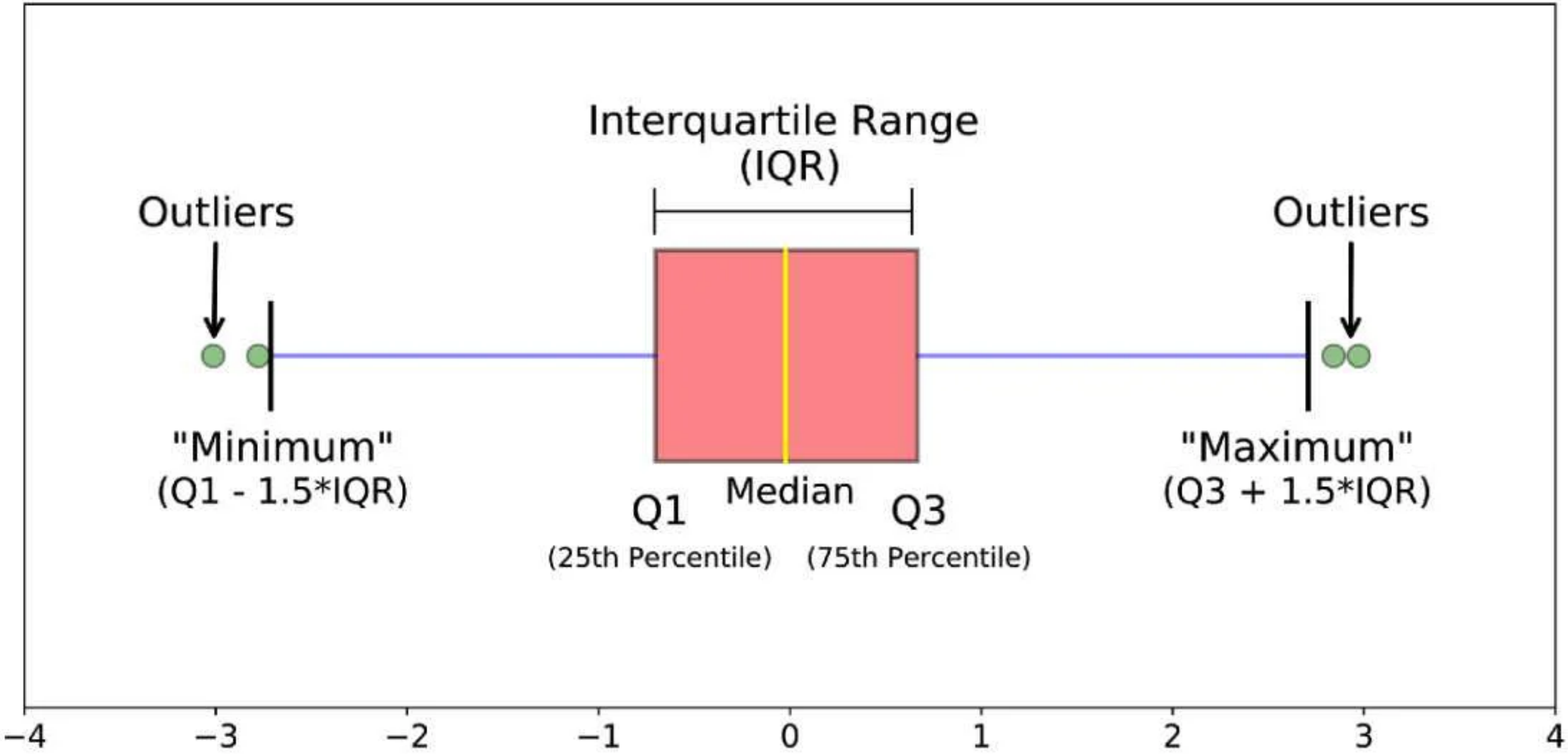


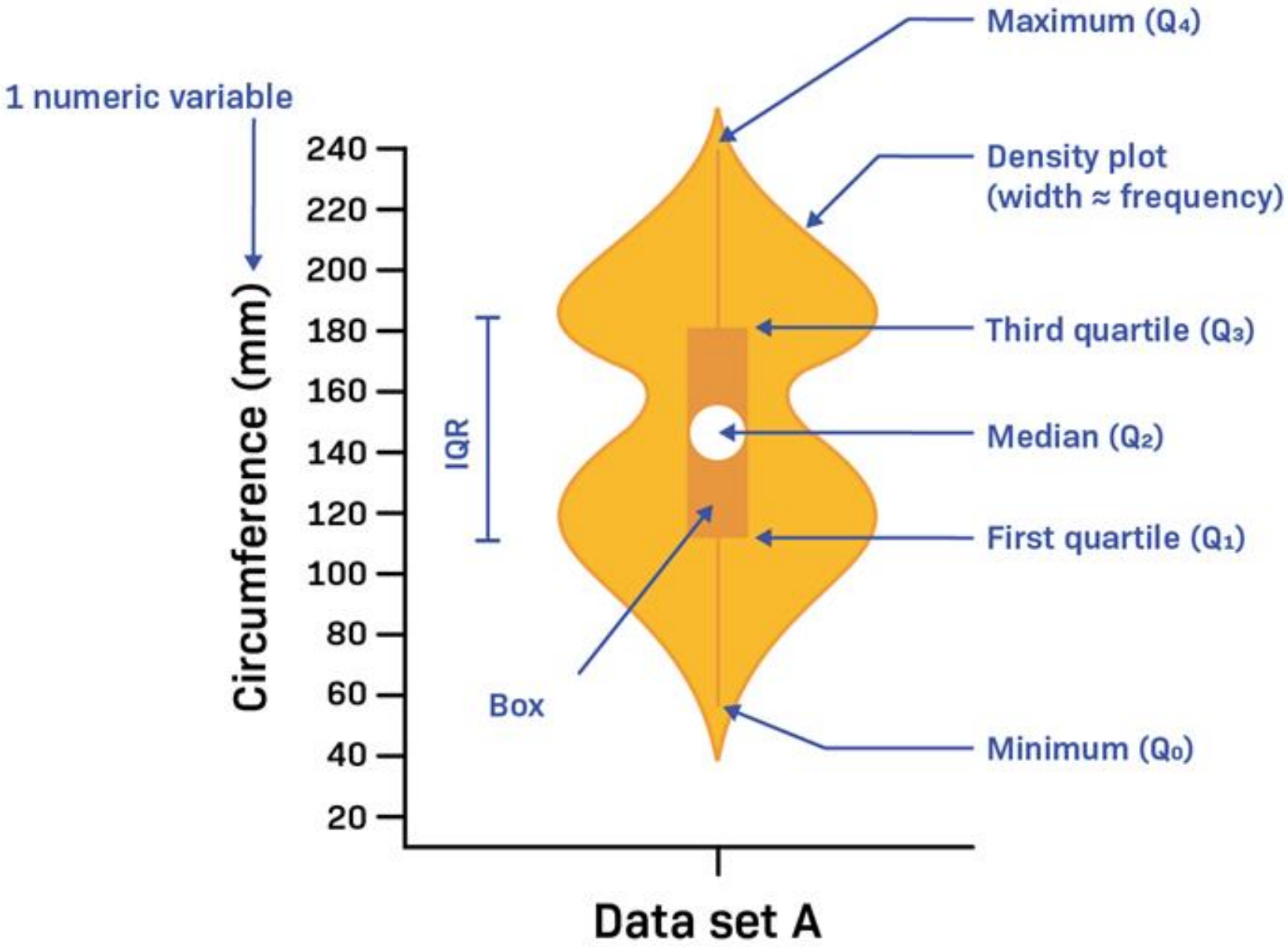
Box Plot Vs Violin Plot




(A)
Frequency
Smooth probability density function (PDF)
Histogram
Circumference (mm)
Rotate 90°
(B)
Frequency
Histogram
Circumference (mm)
Center bars
(C)
Central axis
Circumference (mm)
Estimate a smooth, continuous curve
(D)
Central axis
Smooth PDF
Circumference (mm)
(E)
Central axis
Circumference (mm)


Histogram + KDE
Boxplot
Violin plot
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5


Interquartile Range
(IQR)
Outliers
Outliers
"Minimum"
(Q1 - 1.5*IQR)
Q1
Median
Q3
(25th Percentile)
(75th Percentile)
"Maximum"
(Q3 + 1.5*IQR)
−4
−3
−2
−1
0
1
2
3
4


1 numeric variable
Circumference (mm)
240
220
200
180
160
140
120
100
80
60
40
20
IQR
Box
Maximum (Q4)
Density plot
(width ≈ frequency)
Third quartile (Q3)
Median (Q2)
First quartile (Q1)
Minimum (Q0)
Data set A

Single-Factor Experiments

only a single factor varies

- Fertilizer trials where several rates of a single fertilizer element are tested.
- Insecticide trials where several insecticides are tested.

Two-Factor Experiments

A two variable factors.

- Effect of **Nitrogen** Levels on the growth of **different Wheat varieties**
- To evaluate the effect of **fertilizer type** and **irrigation level** on **wheat yield**

Completely Randomized Design

Treatments are assigned completely at random

Appropriate for homogeneous experimental units

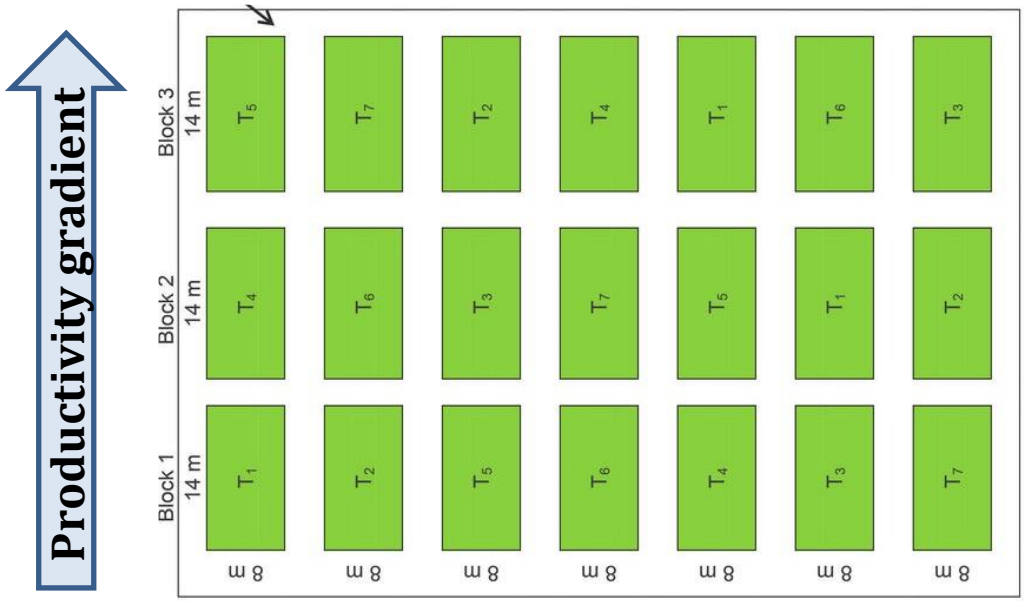
completely randomized design

S4	S2	S3
S3	S4	S1
S3	S2	S4
S1	S2	S1

Randomized Complete Block Design

The experimental area has a predictable productivity gradient.

Suited for field experiments



Variance

The **spread or dispersion** of a set of data points around the mean

Population	Sample
$\sigma^2 = \dfrac{\Sigma(x_i-\mu)^2}{n}$	$S^2 = \dfrac{\Sigma(x_i-\bar{x})^2}{n-1}$
μ - Population Average xi - Individual Population Value n - Total Number of Population σ^2 - Variance of Population	x - Sample Average x_i - Individual Population Value n - Total Number of Sample S^2 - Variance of Sample **Bessel's correction n-1**

ANOVA Table

One-way ANOVA Table

Source of Variance	Degree of Freedom (df)	Sum Square (SS)	Mean Square (MS)	F-ratio
Between Groups (Treatment)	k-1	$SSB = \sum_{j=1}^{k}\left(\frac{T_j^2}{n_j}\right) - \frac{T^2}{n}$ $\quad SSB = \sum_{j=1}^{k} n_j\left(\overline{X}_j - \overline{X}_t\right)^2$	$MSB = \frac{SSB}{k-1}$	$F = \frac{MSB}{MSW}$
Within Groups (Error)	n-k	$SSW = \sum_{j=1}^{K}\sum_{i=1}^{n} X_{ij}^2 - \sum_{j=1}^{K}\left(\frac{T_j^2}{n_j}\right)$ $\quad SSW = \sum_{j=1}^{k}\sum_{i=1}^{n}\left(X_{ij} - \overline{X}_j\right)^2$	$MSW = \frac{SSW}{n-k}$	
Total	n-1	$SST = \sum_{j=1}^{K}\sum_{i=1}^{n} X_{ij}^2 - \frac{T^2}{n}$ $\quad SST = \sum_{j=1}^{k}\sum_{i=1}^{n}\left(X_{ij} - \overline{X}_t\right)^2$		

- SST = SSB + SSW

k: number of groups n: number of samples

df: degree of freedom

Sample ANOVA table

Source of variation	df	PH	LH	CPP	CL	CD	RPC	GPR	DM	AD50	SD50	100SW	KWP	ASI
Replication	2	0.006	0.13	0.02	2.10	0.06	0.28	6.1	0.4	0.13	0.18	1.25	449.7	0.02
Variety	9	1.09***	2.41***	0.18***	69.15***	0.21***	4.76***	214.22***	16.16***	7.90***	17.30***	32.91***	4432.7***	4.00***
Error	18	0.007	0.04	0.03	2.89	0.02	0.29	16.24	1.80	0.05	0.08	3.49	200.6	0.037

Source of variation	df	SS	MS	F
Group	2	343.0821	171.5410	12.8907
Error	325	4324.8947	13.3074	
Total	327	4667.9768	184.8484	

Source	SS (sum of squares)	Df (degree of freedom)	MS(mean square)	F	P-value
Between-groups	21.2564	2	10.6282	7.26683	0.001304
Within-groups	109.6923	75	1.4626		
Total	130.9487	77			

Software Installation